

10 – Opportunities and Risks of AI, Ethical aspects

Carlos Ramos

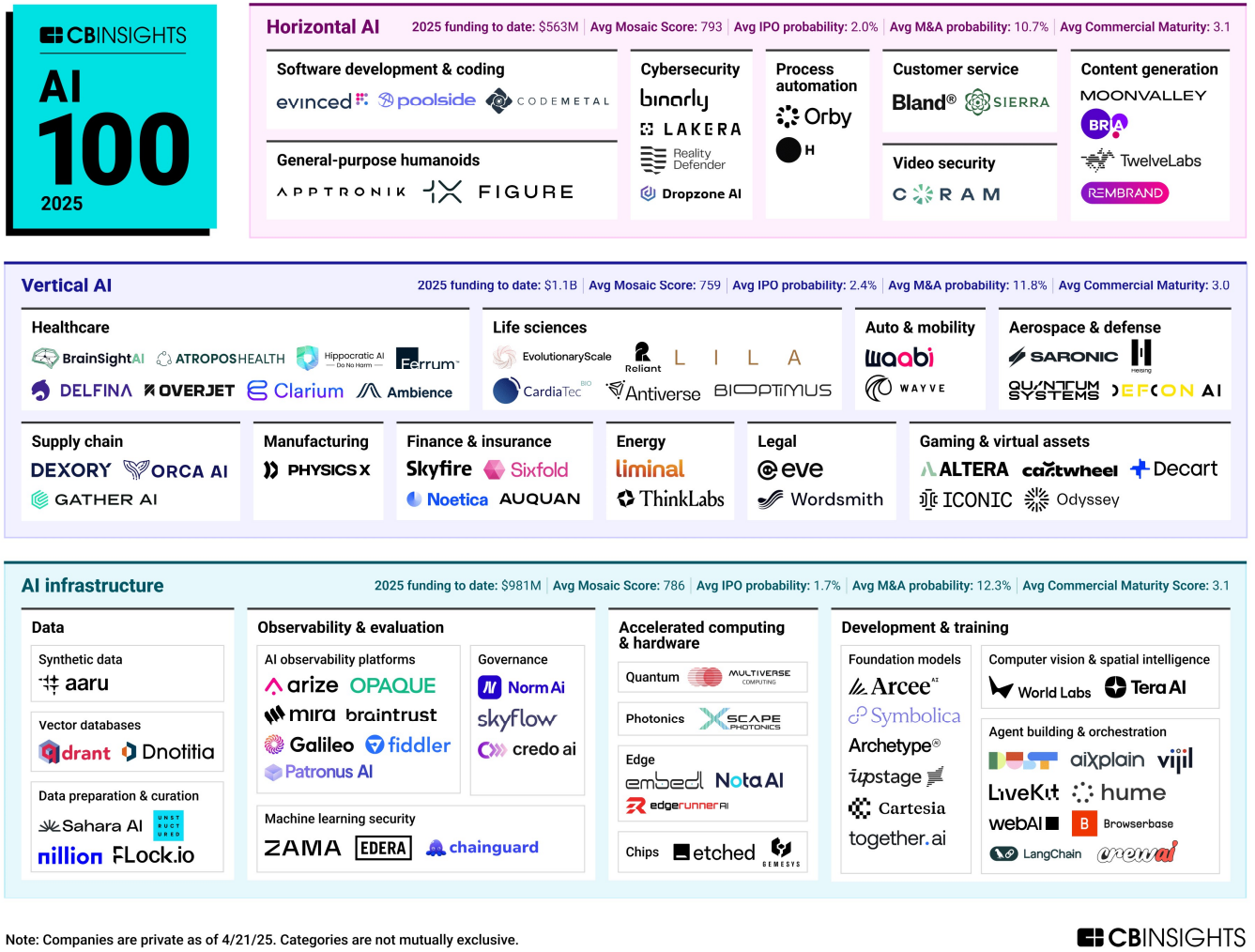
csr@isep.ipp.pt

Curricular Unit: Artificial Intelligence

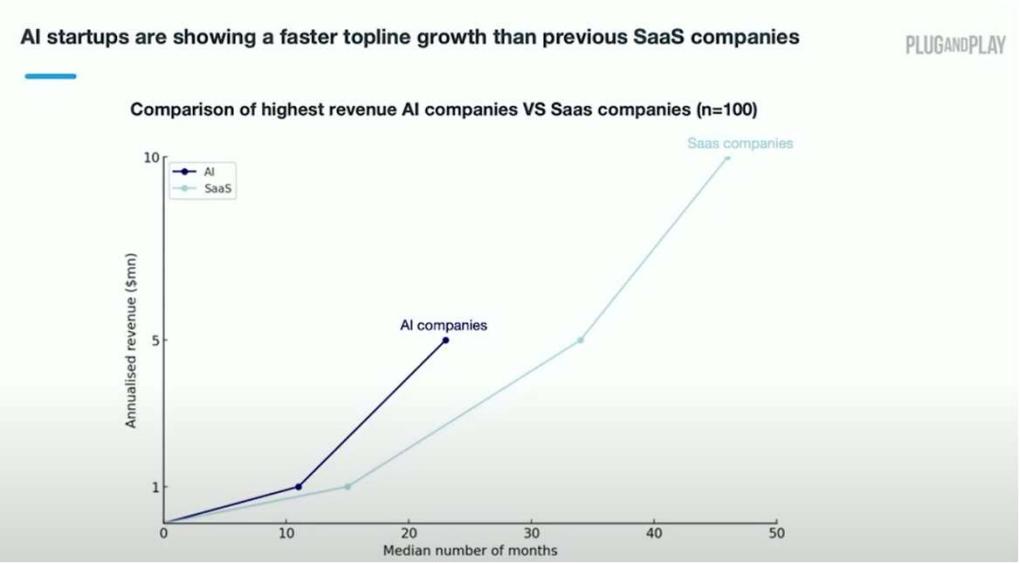
Programme: BSc in Informatics Engineering (ISEP)

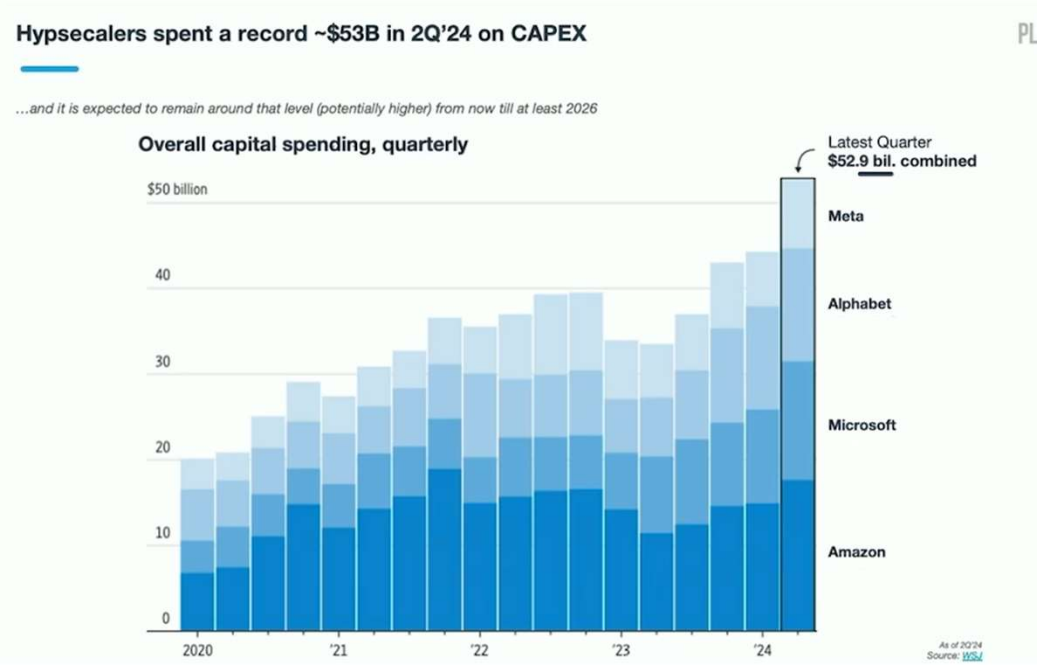
Opportunities of AI

New AI start-ups are appearing every day



Opportunities of AI





CapEx (Capital Expenditure) refers to capital expenditures, that is, the funds a company invests in the acquisition, improvement, or maintenance of long-term physical assets.

https://youtu.be/2V12o6IlfXY

Some Portuguese companies based on AI

Feedzai: founded in 2011, is one of the most powerful artificial intelligence companies in Portugal, specializing in the real-time detection and prevention of financial fraud. The company combines machine learning with advanced data analysis, offering solutions for sectors such as banking, retail, and e-commerce. It became a unicorn in 2021, with a valuation exceeding one billion dollars. It collaborates with renowned global banks such as Citi and BNP Paribas.

Unbabel: is the next-generation LangOps Platform, and all-in-one solution for global translations. The Widn.AI system will allow in three years the need of human intervention in translation, 32 languages are supported.

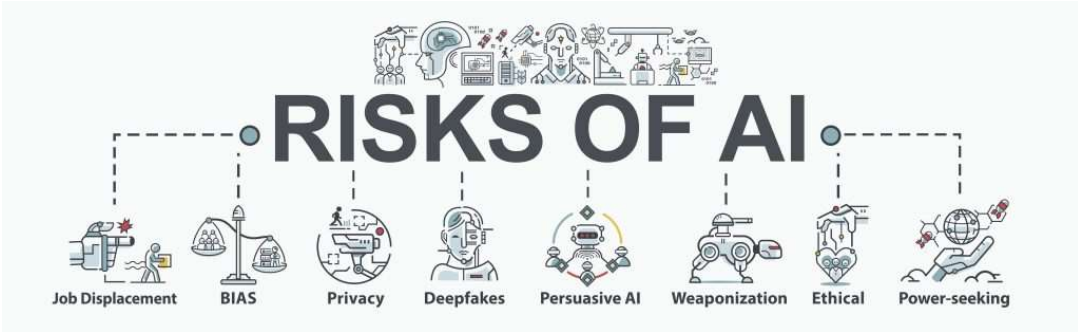
DefinedCrowd: is a company specializing in training machine learning models, with a particular focus on natural language processing (NLP). The company provides high-quality training data for artificial intelligence systems, combining human interaction with AI to improve model accuracy. Recognized as one of the 100 most innovative AI companies in the world.

ADECI: co-founded by Hélder Pereira, MSc in Artificial Intelligence Engineering by ISEP and previously GECAD researcher, is a leading solution in the restaurant and retail market, specializing in sales and product forecasting for multi-establishment and multi-brand businesses. ADECI recently received €500000 in funding from Portugal Ventures.

Risks of AI

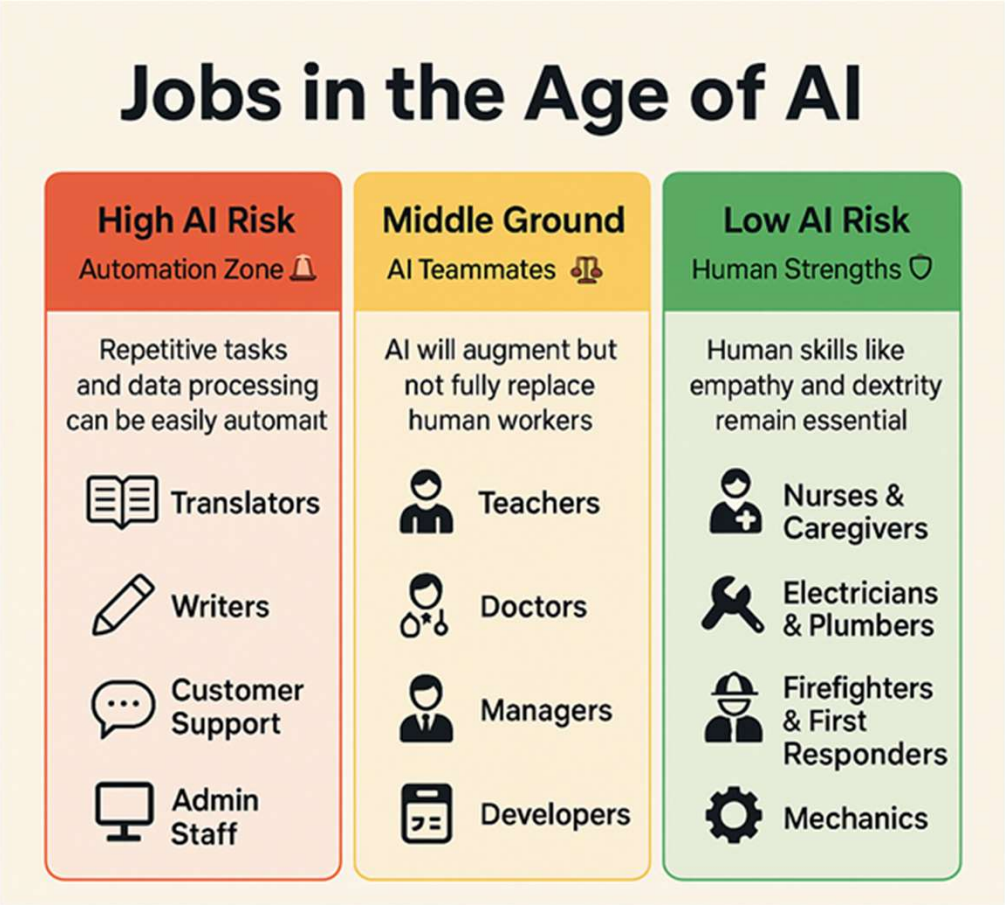
The main challenges of AI stem from their limitations, namely:

- Job Displacement
- Privacy
- Bias
- Environmental Impact
- Fake News
- Security / Cybersecurity
- Cyberwar



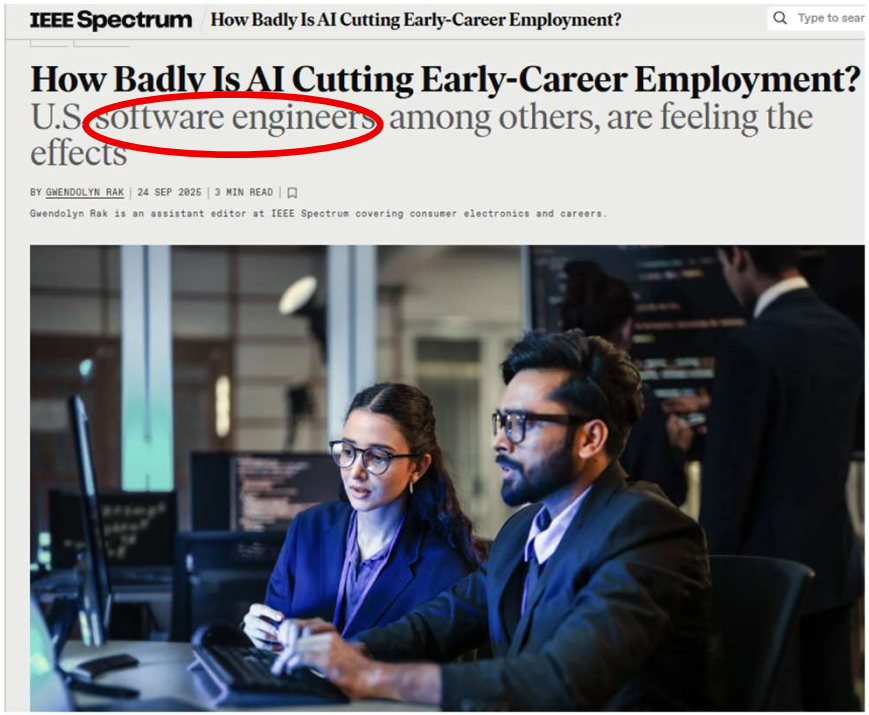
https://it.tufts.edu/costs-and-risks-artificial-intelligence

Job Displacement

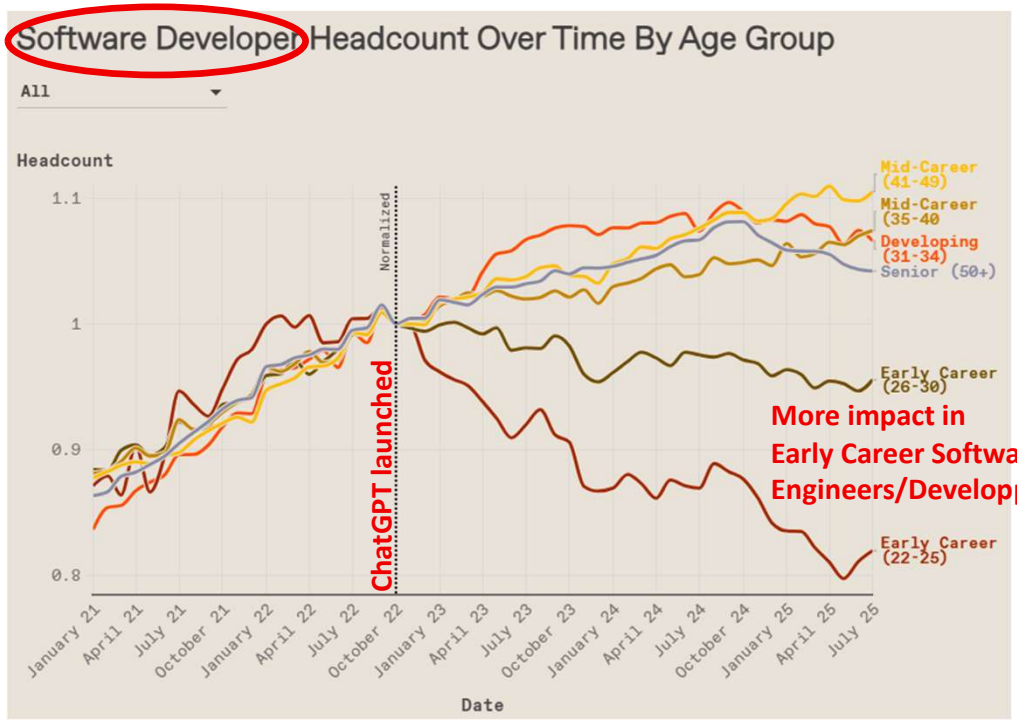




Job Displacement - What about Software Engineers?



https://spectrum.ieee.org/ai-impact-on-job-market



The dotted vertical line is set to November 2022, the month that ChatGPT was released. and the lines are normalized so that they're all equal to 1 in that month.

https://www.understandingai.org/p/new-evidence-strongly-suggest-ai

Job Displacement



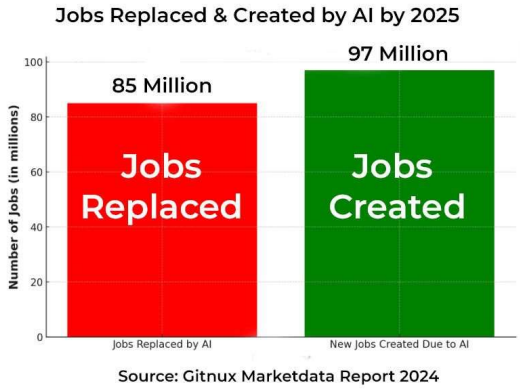
Jobs of Tomorrow: Large Language Models and Jobs

Jobs emerging from the adoption of large language models

WORLD ECONOMIC FORUM

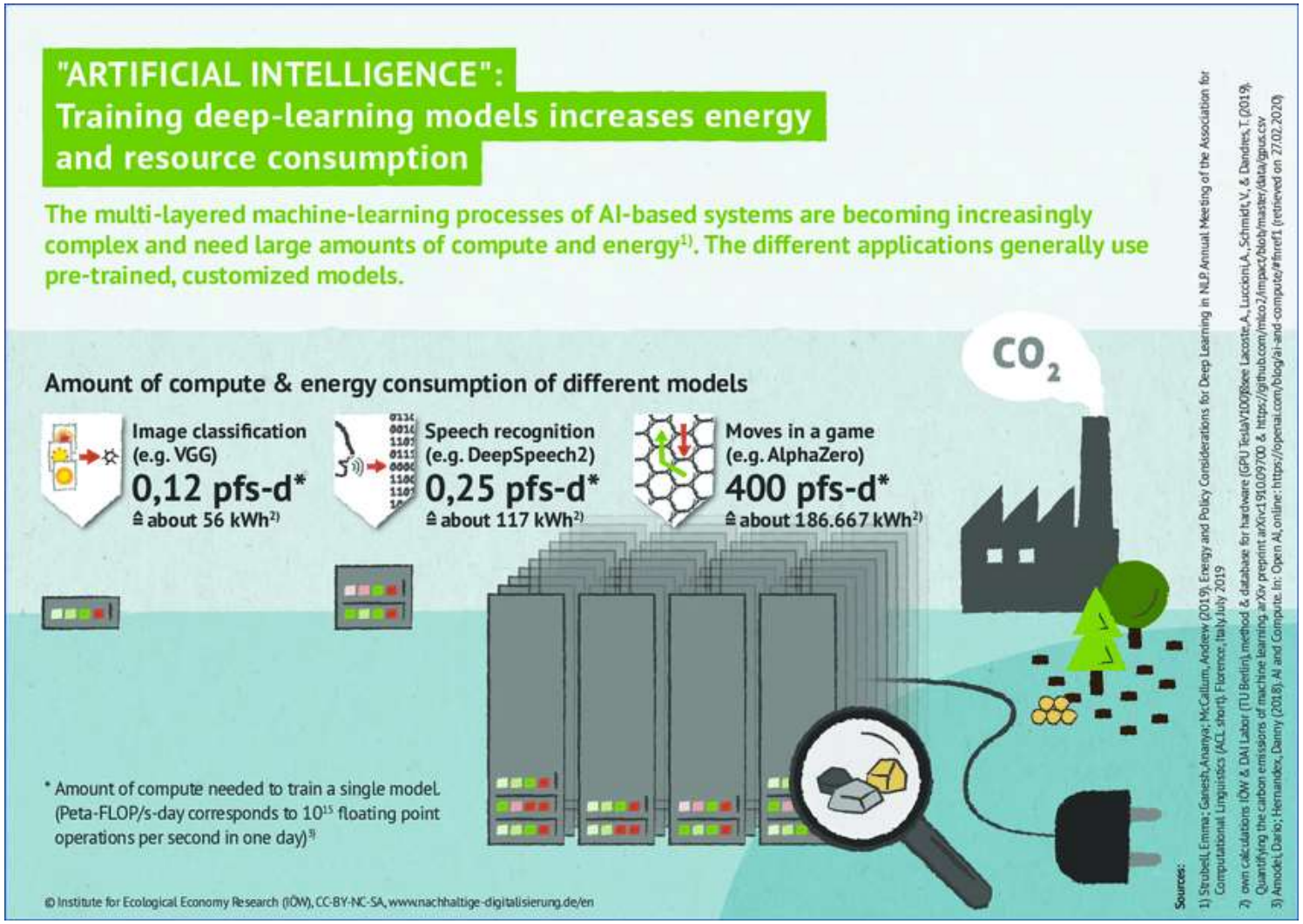
Large language models (LLMs) will transform collaboration between humans and AI, reshaping job roles. While outcomes remain uncertain, potential new job areas could emerge with LLM adoption.

- AI Model and Prompt Engineers
- Interface and Interaction Designers
- AI Content Creators
- Data Curators and Trainers
- Ethics and Governance Specialists



But AI is also creating new jobs!

Environmental Impact



Environmental Impact

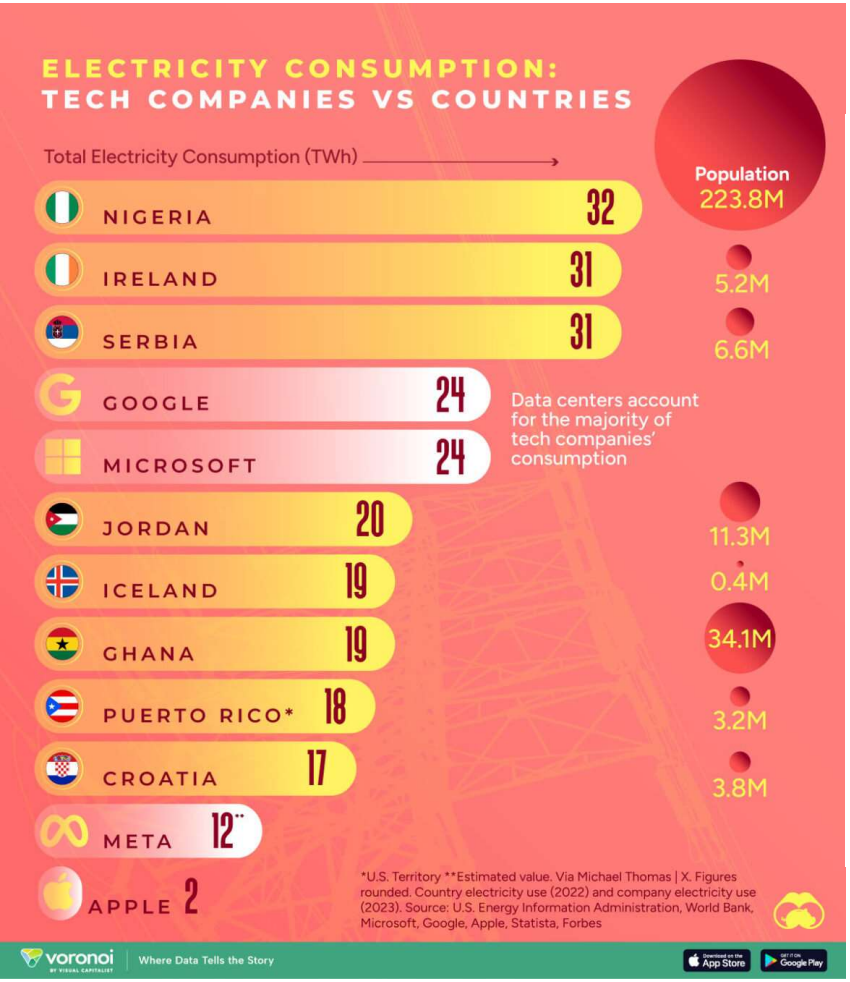
It is estimated that training a model like the Generative Pre-trained Transformer 3 (or GPT-3) consumes about 1,300 megawatt-hours (MWh) of electricity. This is roughly equivalent to the annual energy consumption of 130 homes in the US.

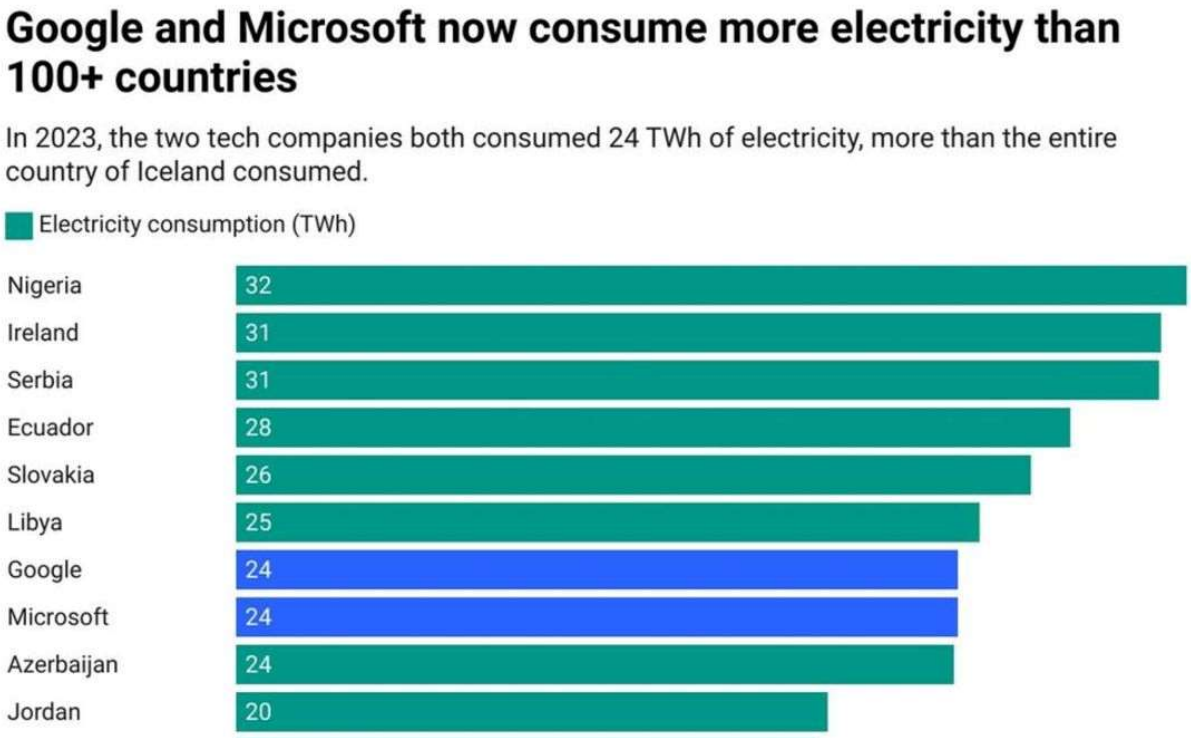
However, it is estimated that training the more advanced GPT-4 consumed 50 times more electricity.

The computing power needed to sustain the growth of AI is roughly doubling every 100 days!

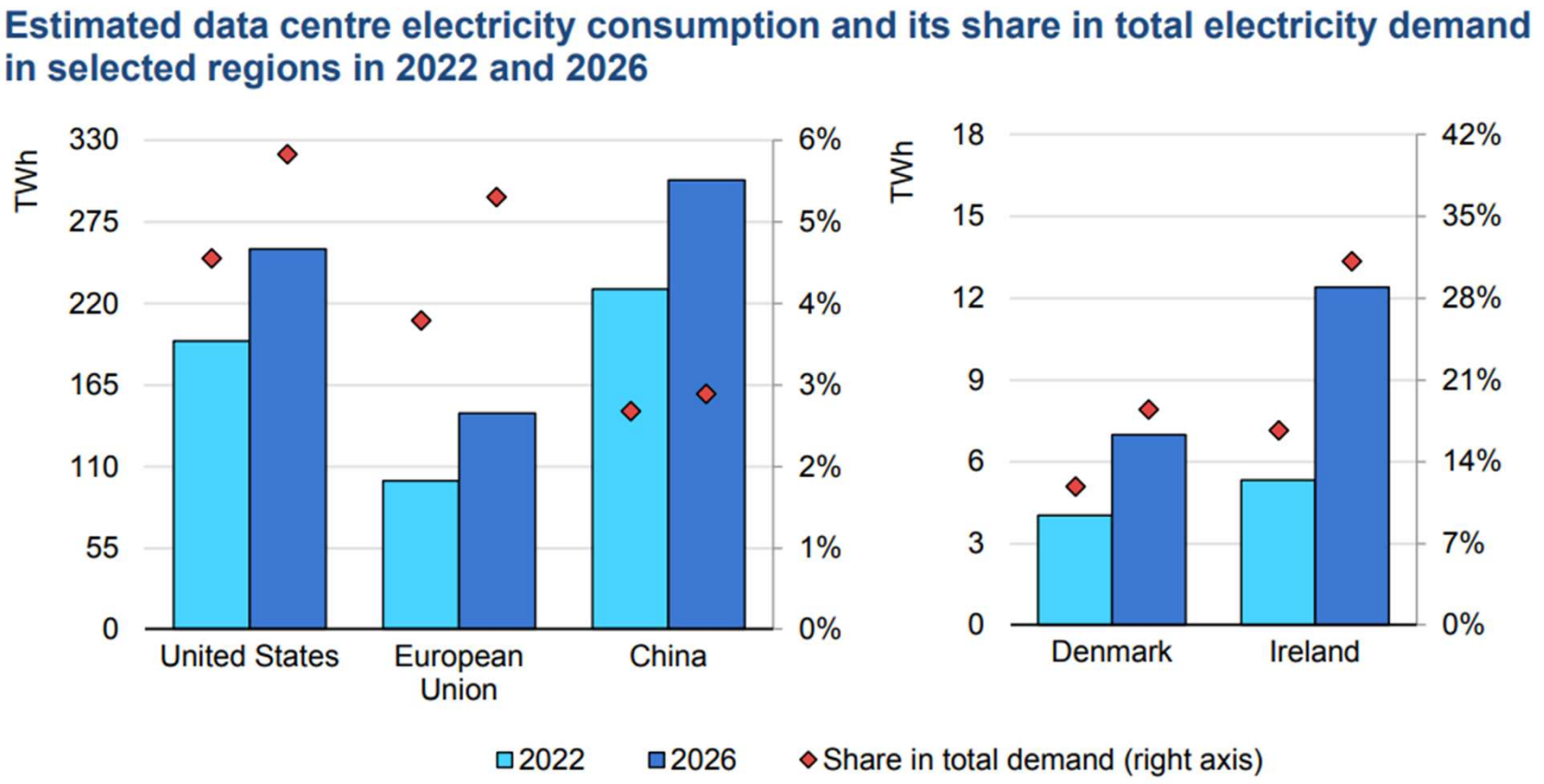
https://www.weforum.org/stories/2024/07/generative-ai-energy-emissions/

Environmental Impact





Environmental Impact



IEA. CC BY 4.0.

Environmental Impact

EUA: central nuclear de Three Mile Island vai reabrir para abastecer a inteligência artificial da Microsoft

A Constellation Energy, proprietária da fábrica na Pensilvânia encerrada desde 2019, planeia voltar a tê-la em funcionamento até 2028, com o gigante tecnológica a comprar toda a energia que produz.

Evan Halper (The Washington Post)
20 de Setembro de 2024, 23:56



A entrada da central nuclear de Three Mile Island, no condado de Dauphin, Pensilvânia, em 2017 CARLO ALLEGRI / REUTERS





Environmental Impact

Green computing refers to the design, manufacture, use, and disposal of computers, chips, other technology components, and peripherals in a way that limits harmful environmental impact, including reducing carbon emissions and energy consumption by manufacturers, data centers, and end users.

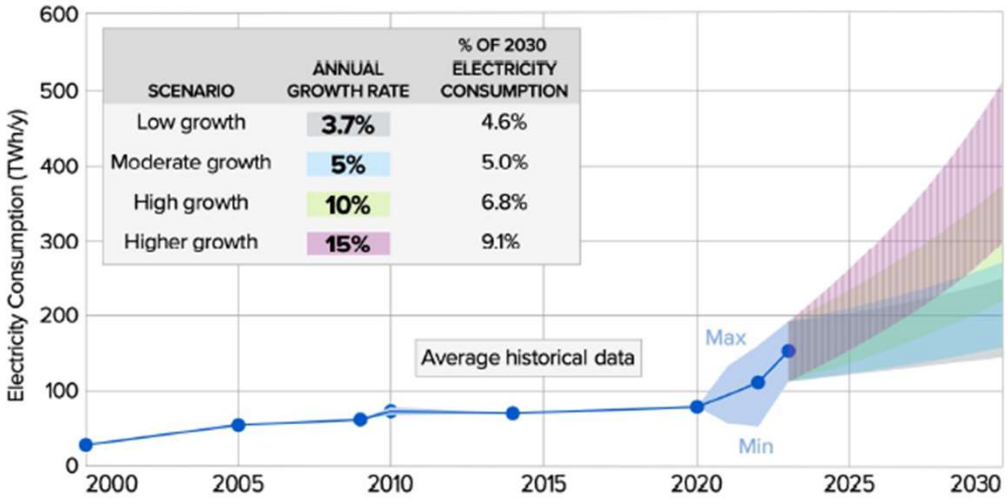
Green computing also encompasses the choice of sustainably sourced raw materials, the reduction of electronic waste, and the promotion of sustainability through the use of renewable resources.

The potential of green computing to have a positive impact on the environment is considerable. The information and communication technologies (ICT) sector is responsible for between 1.8% and 3.9% of global greenhouse gas emissions. The percentage of energy consumption by data centers will be between 4.6% and 9.1% in 2030 compared to total global energy consumption.



1. Invest In Energy Efficient Hardware
2. Virtualization
3. Power Management
4. Invest in Renewable Energy
5. E-Waste Recycling
6. Promote Sustainable Software Development
7. Optimize Your Data Centers

InApp



Environmental Impact

The enormous successes of Generative AI and LLMs, based on Deep Learning models (CNN, RNN and more recently Transformers), have led to a very significant computational effort, especially for training Heavyweight Deep Learning models

This has an impact in terms of time, but also in terms of energy consumption

Some important measures to reduce the energy consumption of Deep Learning systems:

- Use Transfer Learning with pre-trained models, for example YOLO
- Use or combine with alternative approaches that may be lighter (for example Reinforcement Learning)
- Use hardware more suited to Neural Networks, for example NPUs, and reduce the energy consumption of other types of hardware
- Use of Small Language Models (SLM) for specific purposes

Environmental Impact

On the other hand, AI, including models based on ANN and DL, can help reduce energy consumption, making this process more effective and efficient.

For example, based on Demand Response (response to energy demand needs) and Renewable Energy models, we can advise on when certain consumptions should occur, taking advantage of better use of the Energy Grid, especially in the use of renewable energies.

GECAD, ISEP's R&D unit in the area of AI, which has as one of its main areas the application of AI to Energy, has several projects in this regard. Almost all projects use Machine/Deep Learning models, although combining them with other AI technologies such as Optimization, Planning, Multi-Agent Systems and more recently Generative AI and LLMs.

Ethical Aspects

What must be guaranteed:

Justice: It is necessary to ensure that ML/DL/GenAI/LLM algorithms do not discriminate against individuals or groups based on characteristics such as race, gender, or age. Therefore, justice is a fundamental principle in the ethics of this technology, as it seeks to eliminate bias and promote equal treatment.

Transparency: Clear and understandable explanations of how algorithms make decisions must be provided to promote accountability and trust. Transparency is therefore essential to ensure that those affected by algorithmic decisions can understand the factors that influenced those decisions. This can be achieved through a variety of means, such as providing access to the underlying code, documenting the decision-making process, and disclosing the data used to train the algorithm.

Privacy: Privacy has become a fundamental right that must be protected through data protection and ensuring that it is not misused or exploited. Responsible data collection, storage, and use, and the implementation of robust security measures to prevent unauthorized access must be advocated.

Accountability: Holding developers and users of ML/DL/GenAI/LLM systems accountable for their actions and the negative outcomes their systems may generate is necessary to ensure that these algorithms are used in a way that complies with ethical principles, moral responsibility, and social values. This can be achieved through clear guidelines, standards, and oversight mechanisms.

AI Privacy

AI privacy is the practice of protecting personal or sensitive information collected, used, shared or stored by AI

AI privacy is closely linked to data privacy. Data privacy, also known as information privacy, is the principle that a person should have control over their personal data. This control includes the ability to decide how organizations collect, store and use their data. But the concept of data privacy predates AI and how people think of data privacy has evolved with the advent of AI

We can often trace AI privacy concerns to issues regarding data collection, cybersecurity, model design and governance. Such AI privacy risks include:

- Collection of sensitive data
- Collection of data without consent
- Use of data without permission
- Unchecked surveillance and bias (the occurrence of biased results due to human biases that skew the original training data of the AI system)
- Data exfiltration (data theft from AI systems to obtain data or information from the data used for training these AI systems)
- Data leakage (accidental exposure of sensitive data by AI systems)

https://www.ibm.com/think/insights/ai-privacy

Bias

Example 1: Study Reveals Gender Bias in AI Systems for Social Care Decisions

A report in *BMC Medical Informatics and Decision Making* (August 11, 2025) mentions how LSE researchers tested AI-generated summaries of 617 real social care case notes and found gender-related AI bias. They altered only the gender, and ran these cse notes through multiple large language models (LLMs), producing 29,616 comparison pairs. Google's AI tool **Gemma** described men's health issues with terms like "disabled," "unable," and "complex" significantly more often than women's, who were often framed as more independent despite similar needs, an alarming gender bias trend.

Example 2: AI Penalizes Black Women's Natural Hairstyles as Less Professional

A recent test (published on August 10, 2025) using major AI tools revealed that when evaluating images, braids and natural Black hairstyles were more likely to receive lower "intelligence" and "professionalism" scores, biases rarely seen with white women's hair.

Example 6: Apple's Gender-Biased AI for Credit Cards

Apple's credit card algorithm (managed by Goldman Sachs) faced scrutiny after it reportedly offered significantly lower credit limits to women compared to their male spouses, even when women had higher credit scores and incomes.

Example 7: AI Bias in Generative AI

Popular generative AI image tools like DALL·E 2 and Stable Diffusion showed significant biases. When asked to generate images for professions such as "CEO" or "engineer," the AI overwhelmingly produced images of white males. Conversely, prompts like "housekeeper" or "nurse" primarily generated images of women or minorities, reflecting stereotypical biases embedded in training data.

In late 2022, the viral Lensa AI app generated stylized avatars based on users' selfies. Users reported gender biases, with female avatars often sexualized or depicted in suggestive poses, regardless of the original image provided. Male avatars, meanwhile, were more frequently portrayed as heroic or powerful, highlighting stereotypical portrayals of gender

Example 8: LinkedIn's AI-Bias Algorithm in Job Searches

LinkedIn's AI-driven job recommendation systems faced allegations of perpetuating gender biases. A 2022 study introduced a fairness metric to detect algorithmic bias, revealing that LinkedIn's algorithms favored male candidates over equally qualified female counterparts, leading to unequal job recommendations.

Example 9: Facebook's Age-Related AI Bias

ProPublica revealed significant age bias in Facebook's targeted job advertising. Employers were able to exclude older workers from viewing job listings by restricting ad visibility to younger age groups, primarily individuals under 40.

https://www.crescendo.ai/blog/ai-bias-examples-mitigation-guide

Misinformation, Fake News

Generative AI, based on DL, has greatly facilitated the possibility of generating fake news, producing texts with biased narratives and very realistic fake images and videos.

Such capabilities have been used to influence elections, wars, generate fear, and influence many people.

The use of AI to detect false content in texts, images, and videos has been an expanding area in scientific research.







Security / Cybersecurity

Cyber-attacks on the security of systems using AI

But also defense of the security of systems using AI

And with the generalization of AI a new concern appears “Attacks to AI based systems”

On the other hand, as AI becomes increasingly important in decisions, especially critical ones, new concerns arise about cyberattacks on such systems

Several incidents of attacks on vehicles, including autonomous vehicles, have been reported (remote control of temperature, audio and braking of a Jeep Cherokee in 2015; Tesla's navigation system in 2019; Tesla's Autopilot system in 2023)

GECAD participates in many European projects in the area of AI applied to Cybersecurity, such as the SATIE project for the security of European airports and the AIDA project for European security in the area of Defence.





AI Act

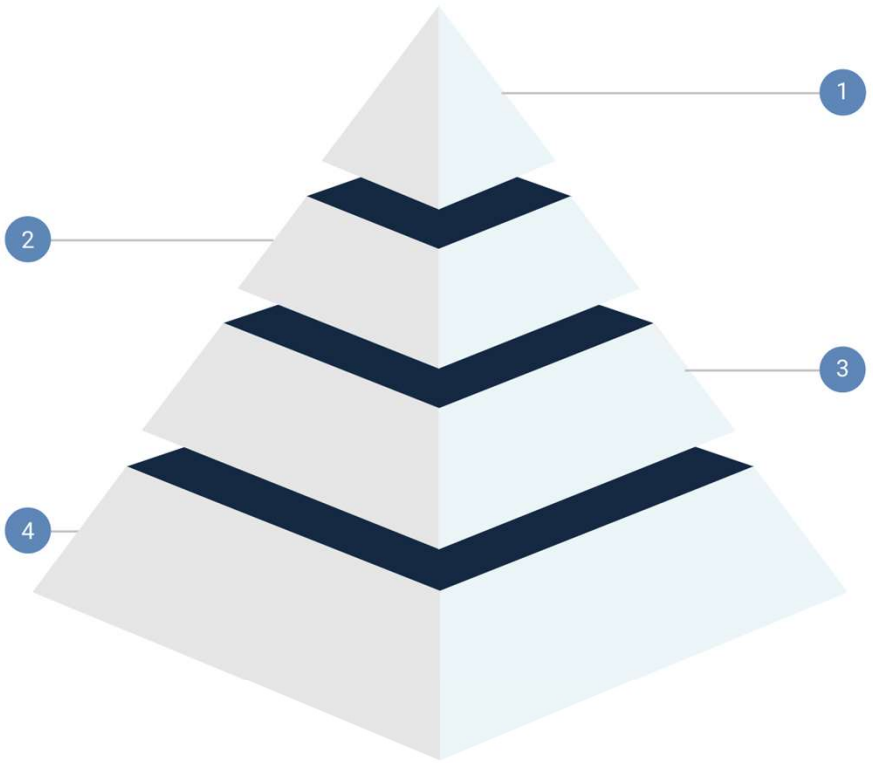
The EU AI Act is the world's first comprehensive law for Artificial Intelligence, setting a risk-based framework for AI systems in Europe, banning unacceptable risks (like social scoring), heavily regulating high-risk AI (e.g., in hiring, critical infrastructure) with strict rules, and imposing transparency for limited-risk AI (like chatbots). It aims to ensure AI is safe, respects fundamental rights, and fosters innovation by creating clear guidelines for developers and users across the AI value chain, potentially becoming a global standard.





https://artificialintelligenceact.eu/

AI Act



Unacceptable Risk

Highest level of risk prohibited in the EU. Includes AI systems using e.g. subliminal manipulation or general social scoring.

Limited Risk

Includes AI systems with a risk of manipulation or deceit, e.g. chatbots or emotion recognition systems. Humans must be informed about their interaction with the AI.

High Risk

Most regulated AI systems, as these have the potential to cause significant harm if they fail or are misused, e.g. if used in law enforcement or recruiting.

Minimal Risk

All other AI systems, e.g. a spam filter, which can be deployed without additional restrictions.

https://www.trail-ml.com/blog/eu-ai-act-how-risk-is-classified

AI Act

Unacceptable risk

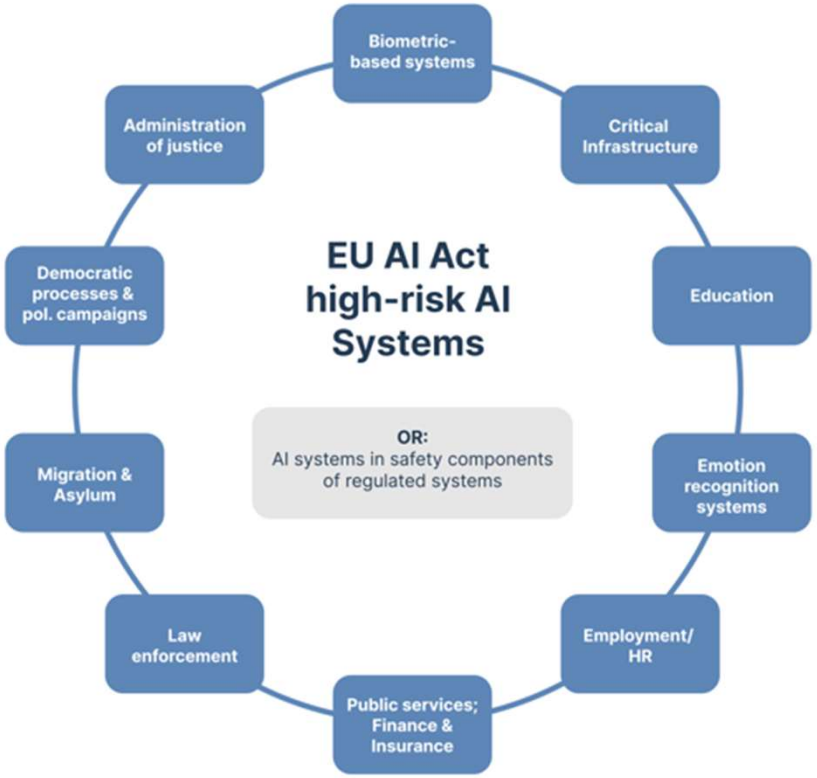
All AI systems considered a clear threat to the safety, livelihoods and rights of people are banned. The **AI Act prohibits eight practices**, namely:

1. harmful AI-based manipulation and deception
2. harmful AI-based exploitation of vulnerabilities
3. social scoring
4. Individual criminal offence risk assessment or prediction
5. untargeted scraping of the internet or CCTV material to create or expand facial recognition databases
6. emotion recognition in workplaces and education institutions
7. biometric categorisation to deduce certain protected characteristics
8. real-time remote biometric identification for law enforcement purposes in publicly accessible spaces

The prohibitions became effective in February 2025. The Commission published two key documents to support the practical application of the prohibited practices:

https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

AI Act





https://www.trail-ml.com/blog/eu-ai-act-how-risk-is-classified

https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai

AI Act

EU proposal to delay parts of its AI Act signal a policy shift that prioritises big tech over fairness

Published: November 24, 2025 6.20pm CET



Critics say the EU is rolling back digital protections under pressure from big tech companies. K_2 Studio

EU AI Act Faces Backlash from Startup Leaders Demanding Implementation Pause

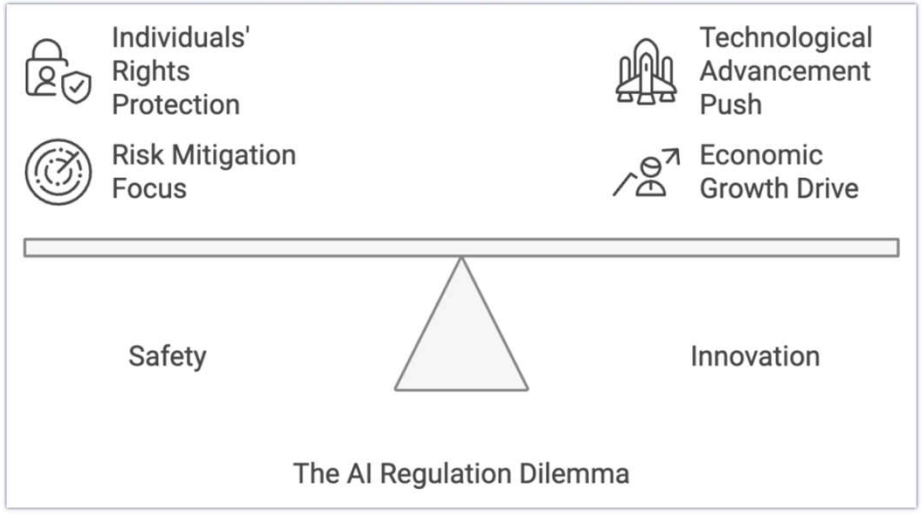
Rosalia Mazza Tuesday, July 1st 2025 04:42 PM



Dozens of Europe's top startups and VCs urge the EU to pause AI Act implementation, warning it risks crushing innovation.

AI Act





The EU AI Act: A Risk-Based, Comprehensive Approach

The EU aims for the AI Act to be an overarching framework, classifying and regulating AI systems according to their societal risks.
At its core is the protection of fundamental rights.
However, the challenge for the EU lies in its regulatory agility.

The U.S.: Guidelines and Narrow Bills

The U.S., by contrast, favors guidelines and sector-specific bills to promote AI safety while leaving room for innovation.
With no comprehensive federal privacy law yet in place, many aspects of AI regulation remain unaddressed.

China AI Regulation: Balancing Development and State control

China's approach may be the most agile, adopting targeted policies to address specific AI issues and enacting laws quickly.
This responsiveness enables China to react to new AI developments and could position it as a model for international AI laws.
However, the potential lack of stability in its regulatory framework may create uncertainty for companies

https://pernot-leplay.com/ai-regulation-china-eu-us-comparison/